

MARIOS VARDALACHAKIS

AI Engineer | Full-Stack Software Engineer | Privacy-Preserving & Trustworthy AI

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PROFESSIONAL SUMMARY

AI and Software Engineer specializing in privacy-preserving, trustworthy AI systems and production-ready ML/LLM architectures. Skilled in bridging AI research with scalable software engineering—building REST APIs, secure data pipelines, and interactive analytics. Proven track record of leading EU-funded R&D projects, deploying secure ML workflows, and publishing peer-reviewed research.

TECHNICAL SKILLS

Machine Learning & AI: Scikit-learn, PyTorch, TensorFlow, Model Training & Evaluation, Feature Engineering, Hyperparameter Tuning, Classification, Clustering, Regression, Predictive Analytics

MLOps & AI Deployment: Docker (basic), CI/CD with GitHub Actions, Model Deployment, API-Based ML Services, Model Serving Pipelines, Evaluation Workflows

Generative AI & LLM Systems: LLM Integration, Prompt Engineering, Retrieval-Augmented Generation (RAG), AI Workflow Orchestration, Agentic AI Concepts, Tool-Calling Architectures, Structured Outputs, LLM-Based Application Design

Software Engineering & APIs: Python, FastAPI, Flask, Django, REST API Design & Integration Pipelines, JavaScript, HTML5, CSS

Data Engineering & Analytics: R, Pandas, NumPy, SQL, PostgreSQL, ETL Pipelines, Statistical Analysis, Secure Data Processing

Data Visualization: Matplotlib, Seaborn, D3.js, dc.js, Interactive Dashboards

PROFESSIONAL EXPERIENCE

AI/ML Full-Stack Engineer & Researcher

2024 – 2026

University of the Aegean — PrivaSI Lab & PRIVACT P.C.

- Designed and deployed **InvisseeAI** (github.com/mariosggg/InvisseeAi), a privacy-preserving ML platform for anonymizing sensitive healthcare datasets, integrating differential privacy, federated learning, and secure multiparty computation.
- Built ML training pipelines and REST APIs to serve the models, linking the data-processing side with the healthcare systems that use them.
- Led a funded Horizon Europe proposal (SECURED Open Call) as Project Lead / Core Developer — responsible for the technical architecture, model design, and delivery, working alongside researchers and engineers on the team.

Administrative & Secretarial Support Staff

2022 – 2023

National Athletic Center of Heraklion (EAKH)

- Provided administrative support to management and staff; managed incoming/outgoing correspondence protocol and record-keeping.
- Worked with partner organizations and public services; welcomed and assisted visitors, and provided basic IT support to staff.

R&D Engineer — Internship & Thesis Project

2016 – 2018

Foundation for Research and Technology – Hellas (FORTH)

- Developed **ShinyAnonymizer** (github.com/mariosggg/ShinyAnonymizer), an R-based platform for anonymizing clinical datasets, with dynamic dashboards for configuring privacy parameters in real time.
- Contributed to the Horizon 2020 iManageCancer project, supporting secure data management tools for cancer-care contexts.

SELECTED PROJECTS

Privacy-Preserving Machine Learning — Case Studies ([GitHub](#) | Python, Scikit-learn, Pandas, NumPy, Matplotlib, PyTorch, Opacus)

- Reproducible ML case studies exploring the trade-offs between data privacy, model utility, and predictive performance.

Sentinel — AI-Assisted Hybrid Threat Intelligence Platform ([GitHub](#) | [Live Demo](#) | HTML, CSS, JavaScript, D3.js, DC.js, Crossfilter.js)

- Developed an explainable analytics platform for investigating suspicious online behavior, coordinated narratives, bot amplification, malicious URLs, and hybrid information threats.

RiskSense — Predictive Intelligence for Early Health Monitoring ([GitHub](#) | [Live Demo](#) | HTML, CSS, JavaScript, D3.js, DC.js, Crossfilter.js)

- Created an interactive analytics dashboard combining environmental and contextual indicators into a confidence-based score for early health monitoring.

IncidentResponse — Cybercrime Complaints & Losses Dashboard ([GitHub](#) | [Live Demo](#) | HTML, CSS, JavaScript, D3.js, DC.js, Crossfilter.js)

- Built an interactive dashboard visualizing global cybercrime complaints and financial losses, with filterable views for complaint trends over time, top countries by complaint volume and losses, and average loss per complaint by country.

IoT Malware Dashboard ([GitHub](#) | [Live Demo](#) | HTML, CSS, JavaScript, D3.js, DC.js, Crossfilter.js)

- Built an interactive dashboard for exploring IoT malware incidents, trends, and statistics in real time.

Epsilon — Ontology Evaluation System ([GitHub](#) | [Live Demo](#) | HTML, CSS, JavaScript, REST API Integration)

- Built an online evaluation system that shows metrics, statistics, and pitfalls of ontologies.

EDUCATION & CREDENTIALS

MPhil by Research in Intelligent Systems (Fully Remote)— University of the Aegean, Department of Cultural Technology and Communication | 2024–2026 | Grade: 9.50/10

M.Sc. in Software Engineering (Fully Remote)— Hellenic Mediterranean University, Department of Electrical and Computer Engineering | 2020–2024 | Grade: 7.50/10

B.Sc. in Informatics Engineering — Hellenic Mediterranean University, Department of Electrical and Computer Engineering | 2011–2020 | Grade: 6.31/10

PUBLICATIONS

- M. Vardalachakis, M. Tampouratzis, and N. Papadakis, “ShinyAnonymizer enhanced version and beyond: A further exploration of privacy-preserving solutions in health data management,” *Appl. Sci.*, vol. 14, no. 16, p. 6921, 2024, doi: 10.3390/app14166921.
- M. Vardalachakis, A. Siatras, and C. Kalloniatis, “InvisseeAI: Advanced healthcare data anonymization platform,” in *Proc. 2025 IEEE Int. Conf. Information, Intelligence, Systems & Applications (IISA)*, 2025, pp. 1–7, doi: 10.1109/IISA66859.2025.11311319.

Plus 6 additional peer-reviewed publications — full list: [Google Scholar](#)